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Linear and non-linear association analysis of genomic ranges annotations

Introduction

With NGS there arise plenty genomic data. Lists of annotated genomic ranges from a biological setting have to be analyzed to answer specific scientific questions. To understand the data it is necessary to analyze the annotations of the genomic intervals as well as to analyze the assoziation between multiple sets of genomic ranges and their annotations. Linear and non-linar correlations have to be taken into account.

Methods

Starting point of our analysis is a set of annotated genomic intervals (chromosom, start, end and columns of further information) from the same genome. Examples include analyzed CHiP-seq data, methylation data or RNA-seq data. In a first step all intervals are linked to the nearest known gene [Figure 1] and the annotations are aggregated in different ways depending on their scale and kind.

Figure 1: Mapping genomic intervals to nearest gene.

In a second step the resulting gene annotations are categorized and summarized according to their scale (binary, nominal, ordinal, interval and rational). To visualize categorical data ordered or unordered barplots are available. For metric data for example histograms or boxplots are suitable. Furthermore, annotations are grouped and compared [Figure 2].

Then an adequate statistic model is chosen to find possible linear and non-linear assoziations between the annotations. We use for this among others correlation coefficients like Pearson correlation, t-test, Wilcoxon-rank sum test, Kruskal-Wallis test and fisher exact test as well as non-linear characteristics like mutual information.

The results are automatically reported in a HTML document to allow the user an easy overview and help to differentiate between significant dependencies and random effects. All pair-wise comparisons are summarized in a heatmap [Figure 3].

For implementation we choose R. In R/Bioconductor annotated genomic ranges can be easily handled as *GRanges* objects [1].

Results

To validate our tool we reanalyzed a published dataset with CHiP-seq, RRBS and RNA-seq data from mouse embryonic stem cells [2]. The paper states that DNA methylation reversibly alters occupancy of histone modification in regulatory regions like promoters and enhancers. We chose a subset of the dataset with CHiP-seq, RRBS data and RNA-data for each wildtype J1, demethylated cells DKO and remethylated cells DKO3a1, DKO3a2, DKO3b1. First all data was linked to known genes with max. distance 100.000 bp and the annotated values were aggregated per gene. Then, all annotations of the genes are summarized individually as well as in groups [Figure 2].

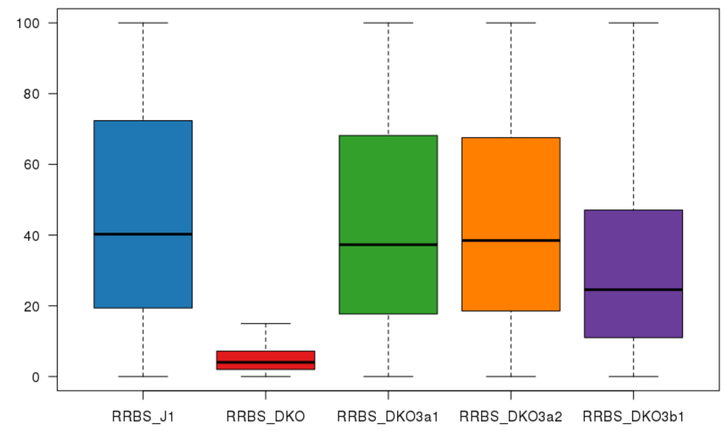

Figure 2: Subset of RRBS data from [2] aggregated per gene.

As summary a heatmap shows the pairwise correlations between the annotations [Figure 3].

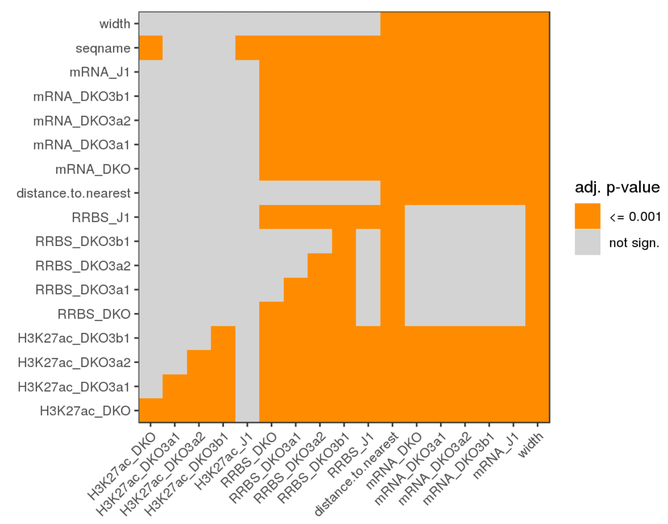

Figure 3: Association overview of dataset [2].

Though p-values have been ajusted to multiple testing using Bonferroni correction the dataset seem to be highly correlated. Furthermore, normalisation to gene could length prevent ambigous impressions.

Outlook / Discussion

As our approach realizes multiple testing, identified dependencies need to be further validated and can only be seen as explorative analysis. Our work includes the comparison of two sets of genomic intervals like the tools StereoGene [3] and GenometriCorr [4] as well as the comparison of an arbitrary number of sets of genomic intervals due to the aggregation on genes. Our future work we will improve the usability of our tool.

References

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